BUS33-4

Labour Economics & Inequalities

Wages, unemployment, and the distribution of income and wealth — read through the causal-inference revolution

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The distributional turn: the same micro and macro toolkit, now forced through the empirical standards modern economics requires.

Albert School — 22 hours

Contents

Preface

You arrive here having built two layers of reasoning. **BUS13-4 Microeconomics** gave you the individual optimiser: a consumer choosing at a tangency, a firm setting MC = MR, a market clearing where supply meets demand, and the income/substitution decomposition of a price change. **BUS23-4 Macroeconomics** gave you the system: aggregate output, the Phillips curve, the NAIRU, and the policy levers that move a whole economy. This course aims that toolkit at one market in particular — the **labour market** — and at one question the first two courses deliberately set aside: not how big the pie is, but **how it is divided**.

That shift is not cosmetic. Labour is the market you will enter directly, and it is the market where economics has been most thoroughly **re-founded** in the last forty years. The reason is causal inference. When micro says a minimum wage "should" cut employment, that is a prediction of a model; whether it actually does is an empirical question, and answering it honestly turns out to be hard, because the world rarely runs the controlled experiment you need. The labour economists who reshaped the field — Card, Angrist and Krueger on minimum wages and schooling; Mortensen and Pissarides on how workers and jobs find each other; Piketty and Milanovic on the long arc of inequality — did so by taking the **identification** problem as seriously as the theory. This course teaches you both halves: the **mechanisms** (supply, demand, search, accumulation) and the **standard of proof** (natural experiments, instrumental variables, difference-in-differences) the field now demands of anyone who claims a policy caused an outcome.

We proceed in the order in which the subject builds. First the two blades of the scissors — labour **supply** (a work–leisure choice) and labour **demand** (a firm's marginal-product condition) — and what **human capital** adds to the supply side. Then the **structure of wages**: why equally productive workers are paid differently, and how to decompose an observed gap. Then **unemployment**: not a failure of the supply-demand diagram but a phenomenon that diagram cannot even express, requiring a model of **search and matching**. Then the **institutions** — minimum wages, unions, insurance — and the Card–Krueger evidence that overturned the textbook prediction. From wages we turn to the **distribution** itself: how to **measure** inequality (Lorenz, Gini) and what drives **wealth** concentration (Piketty's $r > g$). We make the **causal-inference** methods explicit as a toolkit in their own right, and we close on the open frontier — **automation and polarisation** — where we are careful to separate what the evidence can settle from what remains genuinely unknown. As throughout the arc, a concrete number comes before every abstraction, and an honest note on each model's limits comes right after it.

1 Labour supply: the work–leisure choice

The supply of labour is not a primitive; it is the output of a choice every worker makes — how much of a fixed endowment of time to sell. Micro taught you to read any choice as constrained optimisation. Here the good being "bought" with forgone wages is **leisure**, and its price is the **wage**.

1.1 The constrained-optimisation set-up

> **Worked example — the cost of an afternoon off.** You earn €20 an hour. Taking Friday afternoon off — four hours — does not cost you nothing; it costs you €80 of forgone pay. That €80 is the **price of leisure**. If your wage rose to €40, the same afternoon would cost €160: leisure has become twice as expensive, even though nothing about the afternoon changed. Every labour-supply decision is a purchase of leisure at a price equal to the wage.

Definition 1 A worker splits a fixed time endowment T between hours of work h and leisure $\ell = T - h$. With hourly wage w and non-labour income V, consumption is $c = wh + V = w(T - \ell) + V$. The worker solves

$$\max_{c,\ell} u(c, \ell) \quad \text{s.t.} \quad c = w(T - \ell) + V.$$

At the optimum the marginal rate of substitution between leisure and consumption equals the real price of leisure:

$$(\text{MRS})_{\ell,c} = \frac{u_\ell}{u_c} = w.$$

The wage is the slope of the budget line and the opportunity cost of an hour not worked.

1.2 Income and substitution effects of a wage change

This is the micro decomposition of a price change, applied to the price of leisure — and it produces a result with no analogue in ordinary consumer theory.

> **Worked example — why a raise can make you work less.** A nurse earning €25/hour is offered €35/hour. Two pulls act at once. Leisure is now dearer relative to consumption, tempting her to work **more** (substitution). But she is also richer at any given hours, and if leisure is a normal good she will want to consume more of it — to work **less** (income). Which wins is an empirical matter, and at high wages the second routinely dominates: highly paid professionals often cut their hours when paid more. The supply curve **bends backward**.

Definition 2 A rise in w decomposes into two opposing effects on hours:

- **Substitution effect**: leisure becomes more expensive relative to consumption, raising desired hours (always, for a wage rise).
- **Income effect**: the higher wage raises real income; if leisure is normal this raises desired leisure, lowering hours.

The net effect is **ambiguous**. When the income effect dominates, the individual labour-supply curve is **backward-bending**.

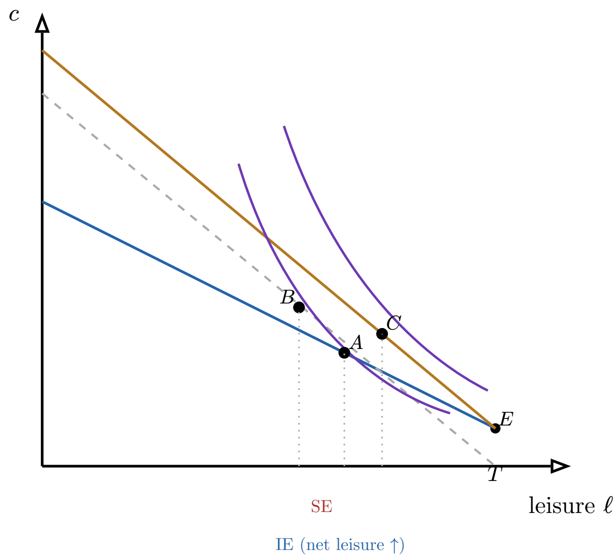

Figure 1: Decomposing a wage rise on the work–leisure diagram. The budget line pivots up around the endowment E (steeper = higher wage). The pure **substitution** effect $A \to B$ — slide along the original indifference curve to the new slope — cuts leisure. The **income** effect $B \to C$ — jump to the higher curve — raises it. Here the income effect wins: net leisure rises ($\ell_C > \ell_A$), so hours fall and supply bends backward.

Common pitfall The substitution effect of a **wage rise** always pushes toward more hours; only the income effect can reverse the sign. So a backward bend is evidence the income effect dominates — never the other way round. And note the asymmetry with ordinary goods: the worker is an **endowment** holder (of time), so a wage change rotates the budget line around E rather than shifting it, and the income effect is large precisely because labour income is most of total income.

2 Labour demand: the marginal-product condition

The other blade of the scissors. A firm does not demand labour for its own sake but for what an extra worker **produces**. This is the firm theory of **BUS13-4**, re-read with labour as the variable input.

2.1 Deriving demand from marginal product

> **Worked example — the last barista.** A café owner asks: should I hire one more barista? The new hire serves €120 of coffee a day (her marginal product valued at the menu price). She costs €100 in wages. Hire her — €20 of surplus. The owner keeps hiring until the **next** barista would add exactly €100, no more. Because each additional worker in a fixed café adds a little less than the last (diminishing returns), there is a unique stopping point, and it falls when wages rise.

Definition 3 A competitive firm with production function $Y = F(K, L)$ hires labour until the **value of the marginal product of labour** equals the wage. With output price p and $\mathrm{MP}_L = \partial F/\partial L$,

$$\underbrace{p \cdot \mathrm{MP}_L}_{\mathrm{VMP}_L} = w, \quad \text{equivalently} \quad \mathrm{MP}_L = \frac{w}{p}.$$

Because MP_L falls as L rises (diminishing returns), the VMP_L schedule slopes **downward**, and it **is** the firm's labour-demand curve: more labour is demanded only at a lower (real) wage.

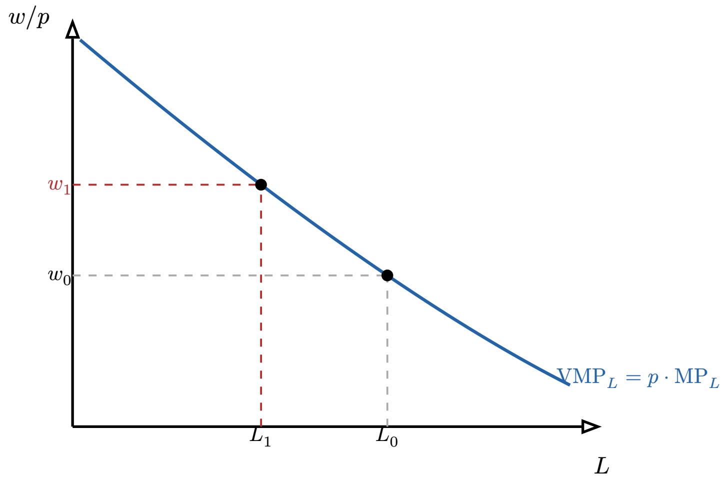

Figure 2: Labour demand is the value-of-marginal-product schedule. A wage rise from w_0 to w_1 moves the firm up the curve, cutting employment from L_0 to L_1. How **much** employment falls depends on the curve's slope — the elasticity of labour demand — which in turn reflects how easily capital substitutes for labour.

2.2 Capital–labour substitution and the automation shock

How far the firm cuts labour when wages rise depends on whether it can swap in **capital**. That swap is governed by one elasticity.

Definition 4 The **elasticity of substitution** σ between capital and labour measures how the input mix responds to the relative factor price w/r:

$$\sigma = \frac{d\ln(K/L)}{d\ln(w/r)} \geq 0.$$

A **large** σ means capital is a close substitute: a wage rise (or a labour-saving technology that cheapens effective capital) triggers a **large** fall in labour demand. A **small** σ means the two are complements and employment barely responds.

Remark A **labour-saving automation shock** enters exactly here. A cheaper, more capable machine lowers the effective price of capital, r, raising w/r. If σ is large the firm substitutes machines for workers and labour demand contracts; if the technology instead raises MP_L (machines that make workers more productive), labour demand can **rise**. Whether automation destroys or creates jobs is, at root, a question about σ and about which margin the technology hits — the theme we return to in the final chapter.

3 Human capital

Workers are not interchangeable units of "labour"; they carry skills, and skills are **produced** by costly investment. Becker's insight was to treat schooling and training with the same present-value logic a firm uses for a machine.

3.1 Becker's model and the rate of return to education

Worked example — is the degree worth it?. A student weighs a three-year degree. It costs €15 000 a year in fees and forgone wages — €45 000 over three years. In return she expects to earn €6 000 more per year for a 40-year career. Is it worth it? Not by comparing €45 000 with $40 \times$ €6000 = €240000 — euros arrive at different dates and must be discounted. The decision turns on the **rate of return** that equates the two present values.

Definition 5 Becker treats education as investment: incur costs C now (direct costs plus forgone earnings) for a stream of extra earnings Δw_t later. The **internal rate of return** r^* solves

$$C = \sum_{t=1}^{N} \frac{\Delta w_t}{(1+r^*)^t}.$$

Education is worth undertaking when r^* exceeds the worker's cost of funds — the same net-present-value rule used for any capital project.

Definition 6 On real earnings data the return is estimated from the **Mincer equation**, a log-linear regression of earnings on years of schooling s and experience:

$$\ln w = \beta_0 + \rho s + \beta_1 \ \mathrm{exp} + \beta_2 \mathrm{exp}^2 + \varepsilon.$$

The coefficient ρ reads as the average proportional earnings gain per extra year of schooling — empirically around 6–10% in many countries.

Common pitfall Reading ρ as the **causal** return to schooling is the field's classic trap. People who get more schooling differ in unobserved ways (**ability**, family, motivation) that also raise earnings — **ability bias** — so the raw correlation overstates the return. Untangling the two is the original problem that drove labour economists toward instrumental variables (Chapter 8): using something that shifts schooling but not earnings directly — compulsory-schooling laws, distance to college, quarter of birth — to recover the causal effect.

4 Wage structure: differentials, discrimination, decomposition

Two workers with the same schooling and experience are often paid differently. Before reaching for "discrimination," an economist must rule out two competing explanations — and then learn to **measure** what is left.

4.1 Compensating differentials (Rosen)

Worked example — why the night shift pays more. Two warehouse workers, identical on paper. One works days at €18/hour; the other works nights, in the cold, at €22/hour. The €4 gap is not unequal treatment — it is the price the firm must pay to staff an unpleasant shift. Strip it out and the jobs pay the same.

Definition 7 **Compensating differentials** are wage differences that offset non-wage job characteristics. Unpleasant, dangerous, or insecure jobs must pay **more** to attract workers; desirable attributes are "paid for" through **lower** wages. Part of any observed wage gap may therefore reflect differences in job **attributes**, not in treatment of **workers**.

4.2 Statistical discrimination

Definition 8 **Statistical discrimination** arises when employers cannot observe an individual's productivity and use **group averages** as a proxy. Members of a group are then paid according to the group mean rather than their own productivity. The result is a persistent group wage gap **without any taste-based prejudice** — and one that can be self-confirming, since the discouraged group may then under-invest in skills the employer cannot see.

4.3 The Oaxaca–Blinder decomposition

Worked example — splitting a 20% gap. Men in a firm earn 20% more than women on average. Of that, suppose differences in measured experience and seniority account for 12 points: if women had the same experience profile, the gap would shrink to 8 points. That residual 8 points — **not** explained by measured characteristics — is what the decomposition isolates for scrutiny.

Definition 9 The **Oaxaca–Blinder decomposition** splits a mean wage gap into an **explained** part (different characteristics) and an **unexplained** part (different returns to those characteristics). Estimate $\ln w = X\beta + \varepsilon$ separately in groups A and B, with mean characteristics $\overline{X}_A, \overline{X}_B$ and coefficients $\hat{\beta}_A, \hat{\beta}_B$:

$$\overline{\ln w}_A - \overline{\ln w}_B = \underbrace{\left(\overline{X}_A - \overline{X}_B\right)\hat{\beta}_B}_{\text{explained (endowments)}} + \underbrace{\overline{X}_A\left(\hat{\beta}_A - \hat{\beta}_B\right)}_{\text{unexplained (returns)}}.$$

Common pitfall The unexplained component is **not** a clean measure of discrimination. It also absorbs any productivity difference you failed to measure and any compensating differential you omitted, and it moves when you add or drop a control. A **statistical-discrimination** reading attributes it to group-based inference; a **compensating-differentials** reading attributes part of it to unmeasured job attributes. The decomposition **frames** the debate; it does not settle it.

5 Unemployment: frictions, matching, and the natural rate

The supply-and-demand diagram has a glaring silence: at the market-clearing wage **everyone who wants a job has one**, so equilibrium unemployment is zero. Reality never is. The resolution is that hiring is not instantaneous — workers and jobs must **find** each other, and that takes time and resources.

5.1 Three kinds of unemployment

Definition 10

- **Frictional**: short-term joblessness from the time it takes to match workers to jobs (search, mobility, labour-market entry). Present even in a booming economy.
- **Structural**: a **mismatch** between the skills or locations of available workers and of available vacancies. Persistent.
- **Cyclical**: a shortfall of jobs caused by deficient aggregate demand over the business cycle (the gap of **BUS23-4**).

5.2 The Mortensen–Pissarides matching model

Worked example — the dating-market view of hiring. A city has 100 000 unemployed workers and 80 000 open vacancies — yet nobody is hired **instantly**. Each day a certain number of workers and firms meet and pair up, like a dating market: more singles on both sides means more matches, but search frictions mean the pool never clears at once. Modelling that meeting process — not a wage that equates two schedules — is the Mortensen–Pissarides contribution.

Definition 11 Search frictions are captured by a **matching function** giving the flow of new hires from the stocks of unemployment U and vacancies V:

$$M = m(U, V),$$

increasing in both and usually constant-returns-to-scale. **Labour-market tightness** is $\theta = V/U$. The job-finding rate is $M/U = m(1, \theta)$ and the vacancy-filling rate is $M/V = m(1/\theta, 1)$. Equilibrium unemployment u equates the inflow (separations) with the outflow (matches): with separation rate δ and the labour force normalised to one,

$$\delta(1 - u) = m(u, V).$$

Vacancy creation is pinned down by **free entry**: firms post vacancies until the expected cost of an unfilled vacancy equals the expected profit from a filled one.

5.3 Reading the Beveridge curve

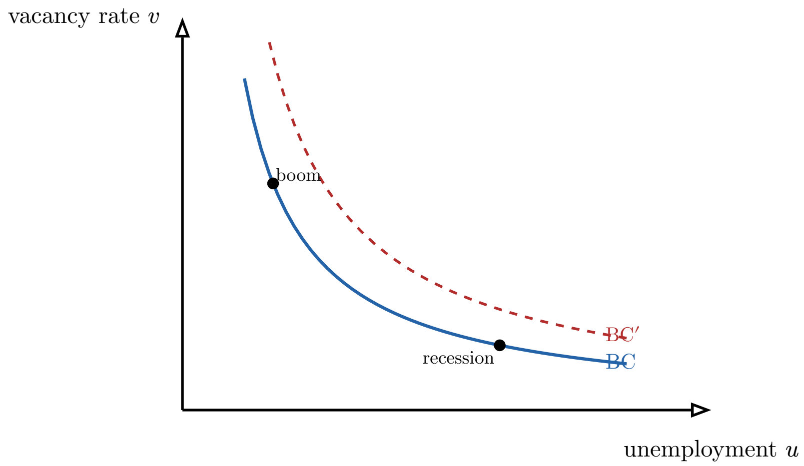

Figure 3: The Beveridge curve relates the vacancy rate to the unemployment rate. Movements **along** it trace the cycle: booms sit upper-left (many vacancies, low u), recessions lower-right. An **outward shift** to BC′ — more unemployment at every vacancy rate — signals worse matching efficiency, i.e. a rise in **structural** unemployment.

Common pitfall Read the **position**, not just the direction. **High vacancies together with high unemployment** is the diagnostic of a **matching/structural** problem — jobs exist but the unemployed cannot fill them. **Low vacancies with high unemployment** points to **deficient demand** (cyclical). Confusing the two leads to the wrong policy: retraining for a structural problem, demand stimulus for a cyclical one.

5.4 The NAIRU

Definition 12 The **NAIRU** (non-accelerating-inflation rate of unemployment) is the equilibrium unemployment rate consistent with **stable** inflation — the labour-market counterpart of the vertical long-run Phillips curve from **BUS23-4**. Frictional and structural unemployment **persist** at the NAIRU; cyclical unemployment is the deviation from it. Driving u below the NAIRU buys accelerating inflation, not a permanent jobs gain.

6 Institutions and the minimum-wage debate

The competitive model makes a sharp prediction about wage-setting institutions — and that prediction is exactly where the empirical revolution struck hardest.

6.1 Minimum wage, unions, and unemployment insurance

Definition 13 In the competitive model a binding wage floor above the market-clearing wage creates a labour surplus (unemployment):

- **Minimum wage**: a binding minimum lifts wages but cuts employment; the size of the cut depends on labour-demand elasticity.

- **Unions**: collective bargaining raises wages above the competitive level for members (the **union wage premium**), which can lower union-sector employment and push workers into the non-union sector.
- **Unemployment insurance**: replacement income during search raises the **reservation wage** and lengthens search, raising frictional unemployment — while insuring consumption and potentially improving match quality.

Remark **Comparative international data** complicate the simple story. Countries with high minimum wages, broad union coverage, and generous insurance (much of continental Europe) do **not** uniformly show higher unemployment than low-institution countries — and the **flexicurity** systems (Denmark) pair generous insurance with light hiring and firing rules and low unemployment. Institutions interact; no single one reads off equilibrium unemployment by itself.

6.2 The Card–Krueger reframing and monopsony

Worked example — New Jersey, 1992. In April 1992 New Jersey raised its minimum wage; neighbouring Pennsylvania did not. Card and Krueger surveyed fast-food restaurants on both sides of the border before and after. The competitive model predicted New Jersey employment should fall relative to Pennsylvania. It did not — if anything, it rose slightly. One carefully designed comparison did more to reshape the field than a decade of theory.

Why might a minimum wage **fail** to cut jobs — or even raise them? Because the competitive model assumes firms are wage-**takers**. If instead an employer has **monopsony** power (it is large relative to its local labour market), it restrains hiring to hold wages down, and a minimum wage can push **both** wage and employment up over a range.

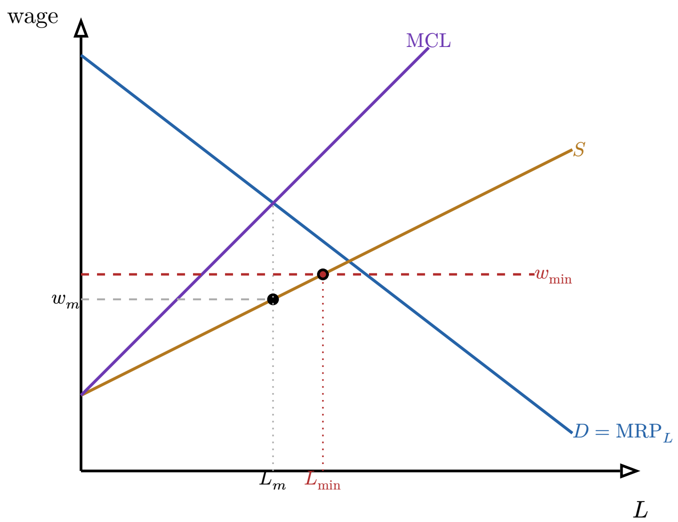

Figure 4: Minimum wage under **monopsony**. The firm equates the marginal cost of labour (MCL, above supply S because hiring one more worker raises the wage paid to all) with demand D, hiring L_m at the low wage w_m. A minimum wage $w_{\min}$ flattens the effective supply: the firm now hires along $w_{\min}$ up to $L_{\min} > L_m$. **Both wage and employment rise** — the result the competitive model cannot produce, and the theory Card–Krueger revived.

Definition 14 What the natural-experiment evidence changed, in three registers:

- **Empirical**: modest minimum-wage rises have small or negligible employment effects, contradicting the naïve competitive prediction.
- **Theoretical**: it revived **monopsony** (employer wage-setting power) as a serious description of low-wage labour markets.
- **Methodological**: it helped install the **natural-experiment** standard now expected of any causal claim in the field — the subject of the next chapter.

Common pitfall The lesson is **not** "minimum wages never cost jobs." It is that the employment effect is an **empirical magnitude**, not a theorem — and that it depends on how far the floor is pushed and on how competitive the local market is. A minimum wage set far above productivity will still destroy jobs; the monopsony range is bounded.

7 Inequality measurement

To study the distribution we first need to **measure** it. Two tools — one visual, one a single number — do almost all the work, and both are built directly from a ranked list of incomes.

7.1 The Lorenz curve

Worked example — five households. Five households earn €10k, €20k, €30k, €40k, €100k — total €200k. Rank them poorest-first and accumulate. The poorest 20% hold €10k = 5% of income; the poorest 40% hold €30k = 15%; the poorest 60%, 30%; 80%, 50%; and 100% hold 100%. Plot those cumulative pairs and you have drawn this society's Lorenz curve — and the sag below the diagonal **is** its inequality.

Definition 15 Rank the population poorest to richest. The **Lorenz curve** $L(p)$ plots the cumulative share of total income held by the poorest fraction p. It runs from $(0, 0)$ to $(1, 1)$, lies on or below the 45° line of perfect equality, and is convex. The farther it sags below the diagonal, the greater the inequality.

7.2 The Gini coefficient

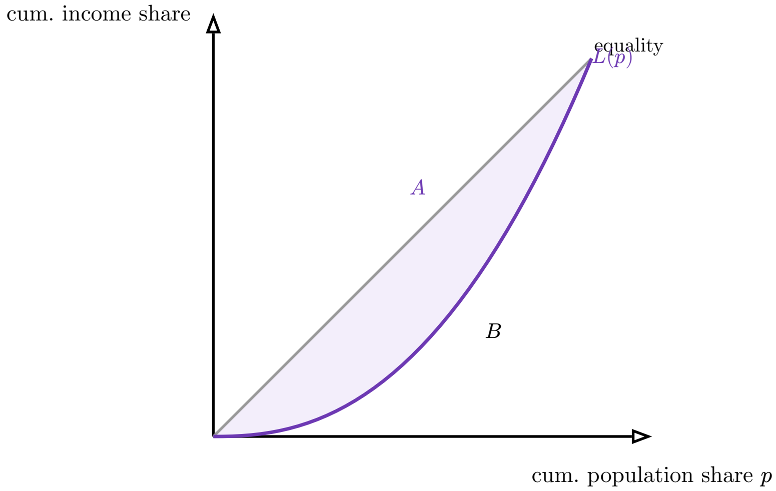

Figure 5: The Gini coefficient is $G = A/(A + B)$, where A is the shaded area between the equality diagonal and the Lorenz curve and B is the area beneath the curve. Since $A + B = 1/2$, $G = 2A$. Perfect equality ($L(p) = p$) gives $A = 0$, $G = 0$; maximal inequality drives $G \to 1$.

Definition 16 The **Gini coefficient** is twice the area between the equality line and the Lorenz curve:

$$G = 2\int_0^1 (p - L(p))\,\mathrm{d}p = 1 - 2\int_0^1 L(p)\,\mathrm{d}p,$$

ranging from 0 (perfect equality) to 1 (perfect inequality). For discrete data with cumulative population shares $p_0 = 0, \ldots, p_n = 1$ and Lorenz ordinates $L_0 = 0, \ldots, L_n = 1$, the trapezoidal approximation is

$$G \approx 1 - \sum_{i=1}^{n} (p_i - p_{i-1})(L_i + L_{i-1}).$$

Remark **The trajectory since 1980.** On real disposable-income data the Gini has **risen** in many advanced economies since around 1980 — most sharply in the United States and United Kingdom, more mildly elsewhere, and barely in some continental and Nordic economies. The dispersion across countries facing the **same** global forces (trade, technology) is itself evidence that **institutions and redistribution** — not technology alone — shape the outcome.

Common pitfall The Gini compresses an entire distribution into one number, so very different distributions can share a Gini. It is relatively insensitive to the **tails** — two societies with the same Gini can differ enormously in their **top 1% share**. For top-end concentration (the Piketty agenda) report top shares alongside the Gini; no single statistic is sufficient.

8 Wealth inequality and its dynamics

Income is a flow; **wealth** is the stock that accumulates from it, and wealth is far more concentrated than income. Piketty's contribution was to give the long-run dynamics of that concentration a single, contestable mechanism.

8.1 The $r > g$ mechanism

Worked example — two families over a generation. The Capital family owns €1m of assets returning $r = 5\%$ a year; they reinvest it. The Labour family earns from work, and wages grow with the economy at $g = 1.5\%$. After a generation the Capitals' fortune has compounded far faster than the Labours' earnings have grown. Inherited wealth pulls away from earned income for no reason other than arithmetic: $r > g$.

Proposition 17 When the (after-tax) rate of return on wealth r exceeds the growth rate of national income g, accumulated and inherited wealth grows faster than incomes from labour. Past wealth is capitalised faster than new income is created, so the capital/income ratio rises and wealth concentrates at the top. In Piketty's framework the steady-state capital/income ratio is

$$\beta = \frac{s}{g},$$

saving rate over growth — so **slower growth raises** the long-run weight of wealth.

Common pitfall $r > g$ is a **tendency**, not a law of motion that runs untouched. It is offset by the dispersion of returns, by wars and crises that destroy capital, by taxation, by saving behaviour, and by how wealth is split across heirs. Piketty's claim is that **absent** such offsets the drift is toward concentration — not that concentration rises monotonically regardless of policy. Read it as identifying the default the rest of the system must work against.

8.2 Intergenerational mobility and the Great Gatsby Curve

Definition 18 **Intergenerational mobility** measures how strongly a child's economic position depends on the parents'. It is summarised by the **intergenerational elasticity of earnings** — the regression coefficient of child on parent log-earnings. A **higher** elasticity means a child's outcome is more tightly bound to the parents', i.e. **lower** mobility.

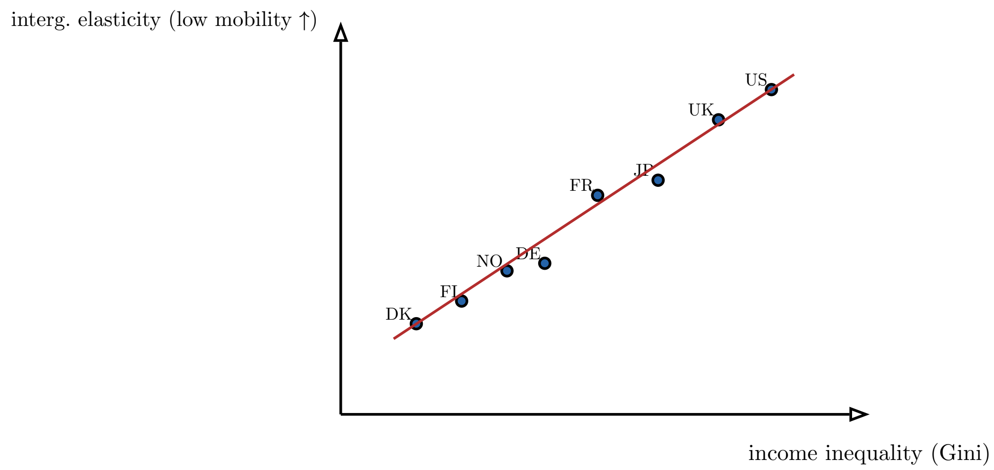

Figure 6: The **Great Gatsby Curve**: across countries, higher income inequality goes with a higher intergenerational elasticity — i.e. **lower** mobility. More unequal societies (US, UK) tend to transmit advantage across generations more strongly than more equal ones (Nordics). Inequality in one generation and immobility across generations reinforce each other. (Positions schematic.)

Common pitfall The Great Gatsby Curve is a **cross-country correlation**, not a proven causal chain. It is consistent with inequality **causing** immobility (the rich buy decisive advantages for their children) but also with common third causes. Stating it as "inequality reduces mobility" without an identification strategy is exactly the inference error the next chapter exists to discipline.

9 Causal inference for labour-market policy

Every chapter so far has bumped into the same wall: a policy and an outcome move together, but does the policy **cause** the outcome? This chapter makes the methods explicit. They are the reason labour economics looks the way it does today, and they are the most transferable skill in the course.

9.1 The counterfactual problem

> **Worked example — the fundamental problem.** A region raises its training subsidy and unemployment falls 2 points. Did the subsidy do it? To know, you would compare the region's actual outcome with what **the same region** would have done **without** the subsidy — but that world is never observed. The missing counterfactual is the whole difficulty; a raw before/after number silently assumes nothing else changed (it always does).

Definition 19 The **causal effect** of a treatment on a unit is the difference between its outcome **with** and **without** treatment — but only one is ever observed. This **fundamental problem of causal inference** means a naïve treated-vs-untreated or before-vs-after comparison **confounds** the policy with every other difference between the comparison groups or periods. All credible methods are devices for **constructing a believable counterfactual**.

9.2 Natural experiments and instrumental variables

Definition 20 A **natural experiment** exploits variation in a policy generated by an external event — a new law, an administrative border, a lottery, a date cutoff — that is **plausibly unrelated** to the outcome's other determinants, so that being treated is "as good as random." Card–Krueger's state border is the canonical case.

Definition 21 An **instrumental variable** Z affects the treatment D but influences the outcome Y **only through** D. The two conditions are **relevance** (Z shifts D) and **exclusion** (Z has no other path to Y). It recovers the causal effect by isolating the part of D that Z drives:

$$\text{effect} = \frac{\text{Cov}(Y, Z)}{\text{Cov}(D, Z)}.$$

Angrist–Krueger's use of **quarter of birth** (which, via compulsory-schooling laws, nudges years of schooling but should not otherwise affect earnings) to estimate the return to education is the textbook example — and a direct answer to the ability-bias problem of Chapter 3.

9.3 Difference-in-differences

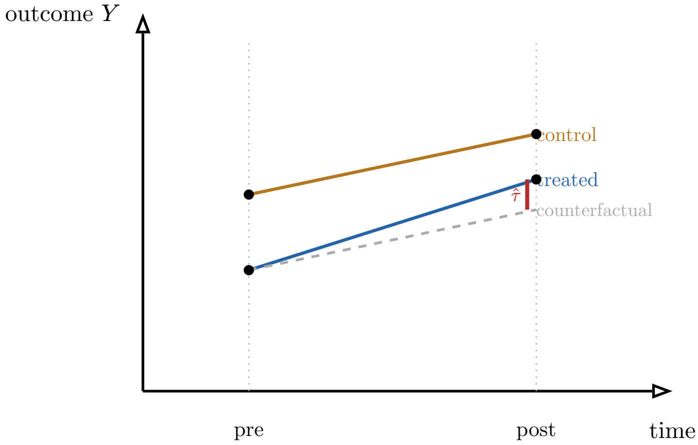

Figure 7: Difference-in-differences. The **control** group's pre-to-post change estimates the common trend. Applying that trend to the treated group's starting point gives the dashed **counterfactual**. The treatment effect $\hat{\tau}$ is the gap between the treated group's **observed** post outcome and that counterfactual — the difference of the two before–after changes.

Definition 22 With a treated group and a control group observed before and after a policy, the **difference-in-differences** estimator is the difference of the two before–after changes:

$$\hat{\tau} = \left(\overline{Y}_{T,\text{post}} - \overline{Y}_{T,\text{pre}}\right) - \left(\overline{Y}_{C,\text{post}} - \overline{Y}_{C,\text{pre}}\right).$$

It nets out fixed differences between the groups **and** any common time trend. Its identifying assumption is **parallel trends**: absent the policy, treated and control would have moved together. Card–Krueger is precisely a DiD design (New Jersey treated, Pennsylvania control).

Common pitfall A causal claim is only as good as its **named identifying assumption**. For DiD that is parallel trends — and it **fails** if something else hit the two groups differently at the same time, or if the groups were already diverging beforehand (check pre-trends!). "Treated and control just differ by X" is not identification until you have argued the counterfactual is credible. Correlation plus a fitted line is not a causal effect.

10 Automation and labour-market polarisation

We close on the live frontier — the question that sends students into this market anxious. Here the discipline of the previous chapter matters most, because the topic is saturated with confident claims the evidence does not support in either direction.

10.1 Routine versus non-routine tasks

Worked example — two jobs, two fates. A payroll clerk and a home-care nurse. The clerk's core tasks are **routine**: codifiable, rule-based, exactly what software does cheaply — so that occupation has shrunk. The nurse's tasks are **non-routine**: physical dexterity, judgement, human contact resist codification — so that occupation has grown. The dividing line that predicts exposure is not "high-skill vs low-skill" but **routine vs non-routine**.

Definition 23 The **task-based view** classifies work by whether its tasks are **routine** (codifiable, rule-based) or **non-routine**, and whether they are **cognitive** or **manual**. Technology **substitutes** most readily for routine tasks (which can be programmed) and **complements** non-routine tasks (raising their productivity and the demand for them).

10.2 Polarisation and complementarity

Definition 24 **Job polarisation** is the empirical hollowing-out of **middle-skill, routine** occupations relative to **high-skill non-routine cognitive** jobs and **low-skill non-routine manual** jobs — a U-shaped change in employment across the skill distribution, documented across the US and Europe since roughly 1980. Where machines **complement** labour, automation raises wages and employment in the affected tasks; where they **substitute**, it displaces workers — the complement/substitute split being the σ of Chapter 2 wearing task-level clothes.

Common pitfall Be scrupulous about **what economics can and cannot adjudicate** here:

- **Can**: that routine tasks are the most exposed to automation, and that the past four decades show measurable polarisation — these are well-identified empirical findings.
- **Cannot (yet)**: the **net long-run effect on total employment and wages**, because automation simultaneously **destroys** tasks and **creates** new ones, and the pace and reach of future technology — including AI moving into non-routine **cognitive** work, the historical safe harbour — are genuinely unknown. A forecast of mass technological unemployment and a forecast of seamless reabsorption are **both** beyond what current evidence can settle. Honesty about that boundary is itself a course learning outcome.

The thread of this book. Labour economics is the micro/macro toolkit aimed at the market you will enter, and forced through a standard of causal proof. **Supply** is a work–leisure choice whose income effect can bend the curve backward; **demand** is the value of marginal product, with capital–labour substitution σ governing the response to wage and automation shocks; **human capital** (Becker, Mincer) makes schooling an investment whose causal return is hard to pin down. The **wage structure** needs compensating differentials and statistical discrimination before any talk of unfairness, and the **Oaxaca–Blinder** decomposition to separate explained from unexplained gaps. **Unemployment** is not a cleared-market artefact but a **search** phenomenon (Mortensen–Pissarides, the Beveridge curve, the NAIRU); **institutions** shift it, and the **Card–Krueger** natural experiment — readable through **monopsony** — overturned the textbook minimum-wage prediction. The **distribution** is measured by the **Lorenz curve** and **Gini**, and driven, for wealth, by Piketty's $\boldsymbol{r > g}$ and its echo in the **Great Gatsby Curve**. Tying it all together is **causal inference** — the counterfactual problem, natural experiments, instrumental variables, and difference-in-differences — the standard by which every claim above must be judged, and the lens through which we read the open **automation** debate: clear about polarisation, honest about what remains unknown.

Exercises

1. A worker has time endowment T, wage w, and non-labour income V. Write the budget constraint, state the tangency condition for optimal leisure, and decompose the effect of a wage rise into income and substitution effects. Explain, with the work–leisure diagram, the conditions under which the individual labour-supply curve bends backward.

2. From a competitive firm's production function, derive the labour-demand curve as the value-of-marginal-product schedule and explain why it slopes downward. Then define the elasticity of substitution σ and use it to predict the effect on employment of (a) a wage rise and (b) a labour-saving automation shock, stating in each case how the answer depends on σ.

3. A degree costs €45 000 (fees plus forgone earnings) and yields €6 000 of extra earnings per year for 40 years. Set up the internal-rate-of-return equation, explain how to decide whether to invest, and then explain why the Mincer-equation coefficient on schooling overstates the **causal** return (ability bias) and how an instrumental variable addresses it.

4. A 20% mean wage gap is observed between two groups. Set up the Oaxaca–Blinder decomposition, identify the explained and unexplained components, and explain why the unexplained component cannot be read directly as discrimination — distinguishing the statistical-discrimination and compensating-differentials interpretations.

5. State the Mortensen–Pissarides matching function and the steady-state condition equating separations to matches. Then read three Beveridge-curve scenarios — high vacancies with high unemployment, low vacancies with high unemployment, and an outward shift of the whole curve — classifying each as frictional, cyclical, or structural and justifying the call.

6. Using the competitive model, explain how a binding minimum wage, a union wage premium, and unemployment insurance each affect equilibrium unemployment. Then explain, with the monopsony diagram, how a minimum wage can raise **both** wage and employment, and state precisely what the Card–Krueger natural-experiment evidence changed — empirically, theoretically, and methodologically.

7. Given five households' incomes, construct the Lorenz curve and compute the Gini coefficient using the trapezoidal formula. Interpret the result, sketch how the curve shifts as inequality rises, and explain why two distributions can share a Gini yet differ sharply in top-1% share.

8. Explain the $r > g$ mechanism and the steady-state capital/income ratio $\beta = s/g$. Relate rising wealth concentration to intergenerational mobility via the Great Gatsby Curve, and explain why the curve is a correlation that does not by itself establish that inequality causes immobility.

9. State the fundamental problem of causal inference. For a specific labour-market policy, set up a difference-in-differences estimator, write it as the difference of two before–after changes, name the parallel-trends identifying assumption, and explain one way it could fail and how you would check for it.

10. Using the routine/non-routine task framework, explain job polarisation and the complement/substitute distinction (relating it to the elasticity of substitution). Then list precisely which claims about automation's effect on employment current economics can adjudicate and which remain genuinely uncertain, and why.